

Principal Component Analysis

Curse of Dimensionality.

- A major problem is *the curse of dimensionality*.
- If the data x lies in high dimensional space, then an enormous amount of data is required to learn distributions or decision rules.
- Example: 50 dimensions. Each dimension has 20 levels. This gives a total of 20^{50} cells. But the no. of data samples will be far less. There will not be enough data samples to learn.

Curse of Dimensionality

- One way to deal with dimensionality is to assume that we know the form of the probability distribution.
- For example, a Gaussian model in N dimensions has $N + N(N-1)/2$ parameters to estimate.
- Requires $O(N^2)$ data to learn reliably. This may be practical.

Dimension Reduction

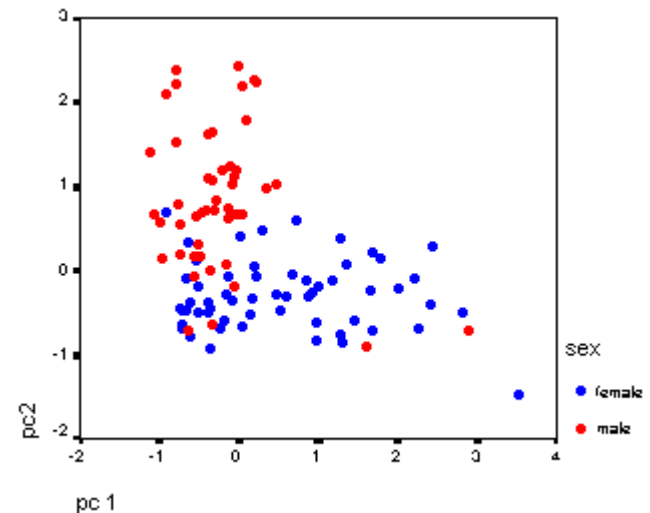
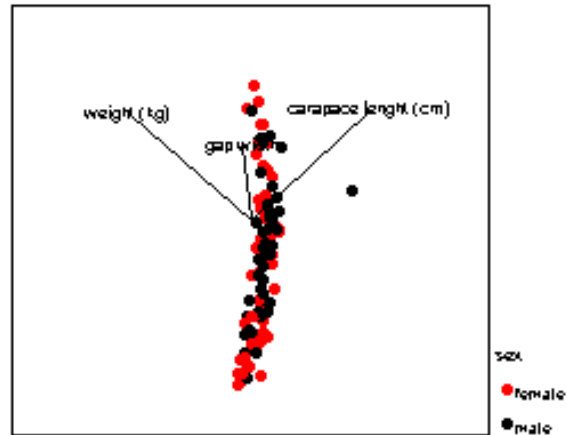
- One way to avoid the curse of dimensionality is by projecting the data onto a lower-dimensional space.
- Techniques for dimension reduction:
- Principal Component Analysis (PCA)
- Fisher's Linear Discriminant Analysis
- Multi-dimensional Scaling.
- Independent Component Analysis (ICA).
- etc.

Philosophy of PCA

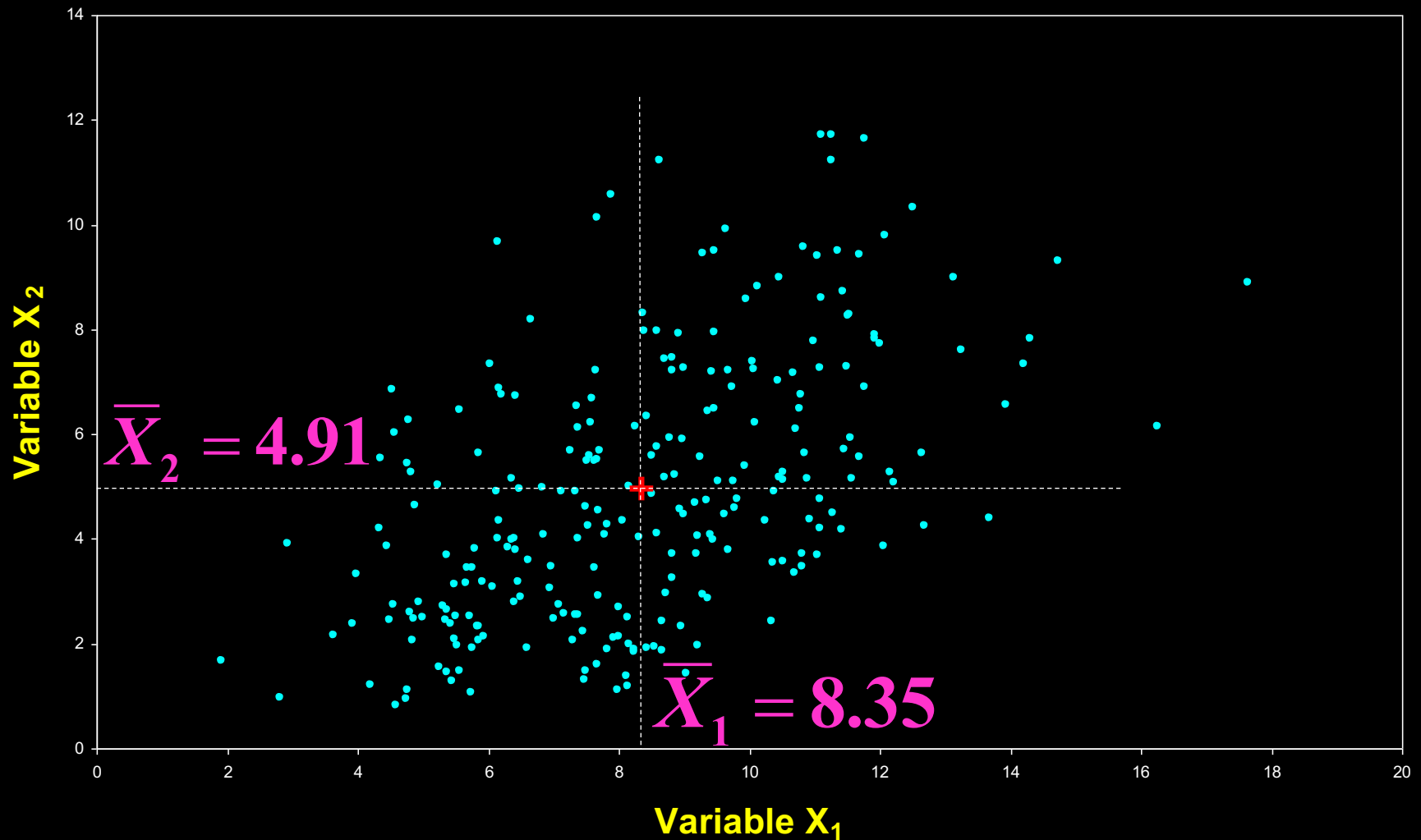
- Introduced by Pearson (1901) and Hotelling (1933) to describe the variation in a set of **multivariate data** in terms of a set of **uncorrelated variables**
- We typically have a data matrix of n observations on p correlated variables $x_1, x_2, \dots, x_p$
- PCA looks for a transformation of the x_i into p new variables y_i that are uncorrelated

The data matrix

case	ht (x_1)	wt(x_2)	age(x_3)	sbp(x_4)	heart rate (x_5)
1	175	1225	25	117	56
2	156	1050	31	122	63
n	202	1350	58	154	67



- 2D Example: variables X_1 and X_2 have positive covariance & each has a similar variance.



$$V_1 = 6.67$$

$$V_2 = 6.24$$

$$C_{1,2} = 3.42 \quad 7$$

Reduce dimension

- The simplest way is to keep one variable and discard all others: not reasonable!
- Weigh all variable equally: not reasonable
- **Weighted average** based on some criterion.
- **Which criterion?**

Let us write it first

- Looking for a transformation of the data matrix $\mathbf{X}$ ($n \times p$) such that



$$Y = \mathbf{a}^T \mathbf{X} = a_1 X_1 + a_2 X_2 + \dots + a_p X_p$$

- Where $\mathbf{a} = (a_1, a_2, \dots, a_p)^T$ is a column vector of weights with

$$a_1^2 + a_2^2 + \dots + a_p^2 = 1$$

One good criterion

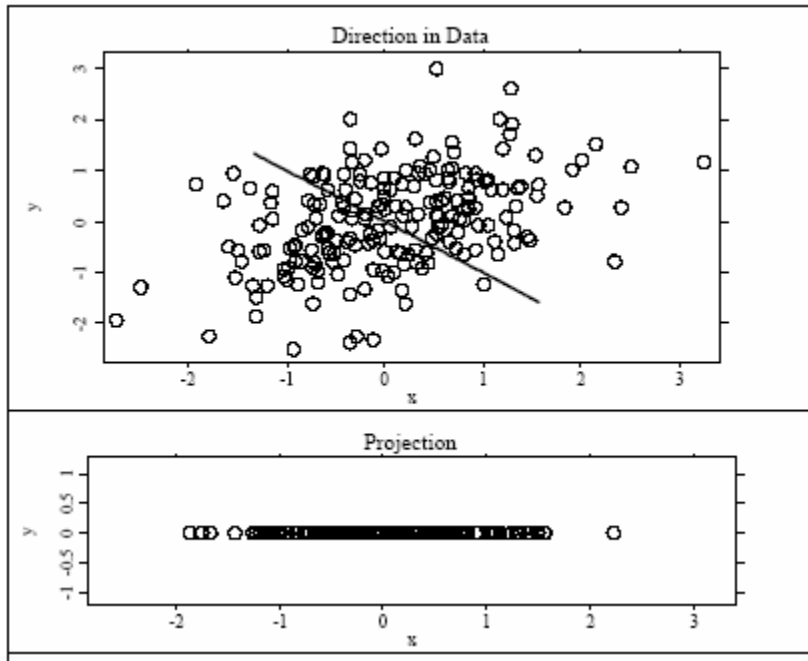
- Maximize the variance of the projection of the observations on the Y variables
- Find $\mathbf{a}$ so that

$$\text{Var}(\mathbf{a}^T \mathbf{X}) = \mathbf{a}^T \mathbf{Var}(\mathbf{X}) \mathbf{a} \quad \text{is maximal}$$

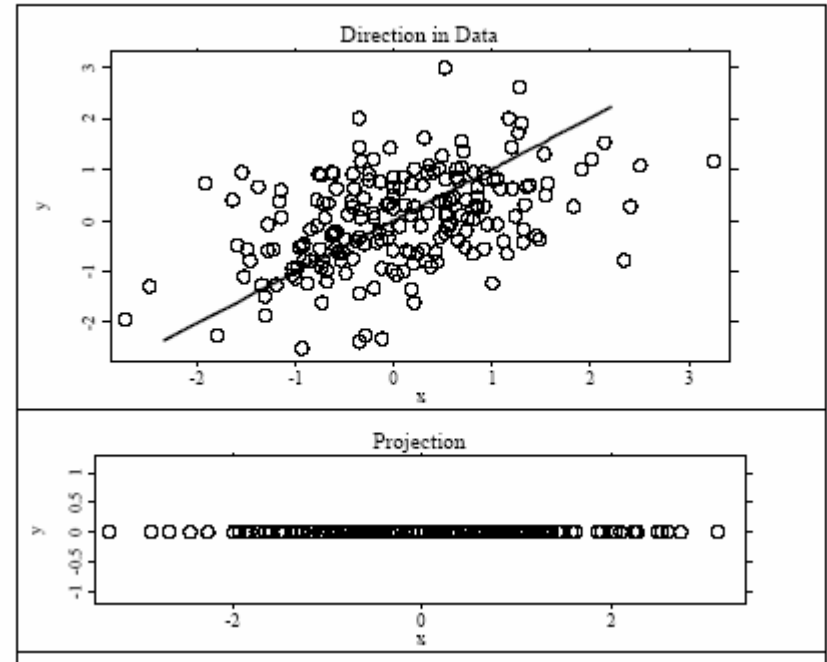
- The matrix $\mathbf{C} = \mathbf{Var}(\mathbf{X})$ is the covariance matrix of the X_i variables

Let us see it on a figure

Good



Better



Covariance matrix

$$C = \begin{pmatrix} v(x_1) & c(x_1, x_2) & \dots & c(x_1, x_p) \\ c(x_1, x_2) & v(x_2) & \dots & c(x_2, x_p) \\ \vdots & \vdots & \ddots & \vdots \\ c(x_1, x_p) & c(x_2, x_p) & \dots & v(x_p) \end{pmatrix}$$

And so.. We find that

- The direction of $\mathbf{a}$ is given by the eigenvector γ_1 corresponding to the largest eigenvalue of matrix $\mathbf{C}$
- The second vector that is orthogonal (uncorrelated) to the first is the one that has the second highest variance which comes to be the eigenvector corresponding to the second eigenvalue
- And so on ...

So PCA gives

- New variables Y_i that are linear combination of the original variables (x_i):
- $Y_i = a_{i1}x_1 + a_{i2}x_2 + \dots + a_{ip}x_p$; $i=1..p$
- The new variables Y_i are derived in **decreasing order of importance**;
- they are called the '**principal components**'

Calculating eigenvalues and eigenvectors

- The eigenvalues λ_i are found by solving the equation

$$\det(C - \lambda I) = 0$$

- Eigenvectors are columns of the matrix A such that

$$C = A D A^T$$

- Where

$$D = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ 0 & & & \\ 0 & \dots & \dots & \lambda_p \end{pmatrix}$$

An example

- Let us take two variables with covariance $c > 0$
- $\mathbf{C} = \begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix}$ $\mathbf{C} - \lambda \mathbf{I} = \begin{pmatrix} 1 - \lambda & c \\ c & 1 - \lambda \end{pmatrix}$

$$\det(\mathbf{C} - \lambda \mathbf{I}) = (1 - \lambda)^2 - c^2$$

- Solving this we find $\lambda_1 = 1 + c$
 $\lambda_2 = 1 - c < \lambda_1$

and eigenvectors

- Any eigenvector A satisfies the condition

$$CA = \lambda A$$

$$A = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad CA = \begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} a_1 + ca_2 \\ ca_1 + a_2 \end{pmatrix} = \begin{pmatrix} \lambda a_1 \\ \lambda a_2 \end{pmatrix}$$

Note: make use of $a_1^2 + a_2^2 = 1$

Solving it we find $A_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$, $A_2 = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$

PCA is sensitive to scale

- If you multiply one variable by a scalar you get different results (can you show it?)
- This is because it uses covariance matrix (and not correlation)
- PCA should be applied on data that have approximately the same scale in each variable

Interpretation of PCA

- The new variables (PCs) have a variance equal to their corresponding eigenvalue

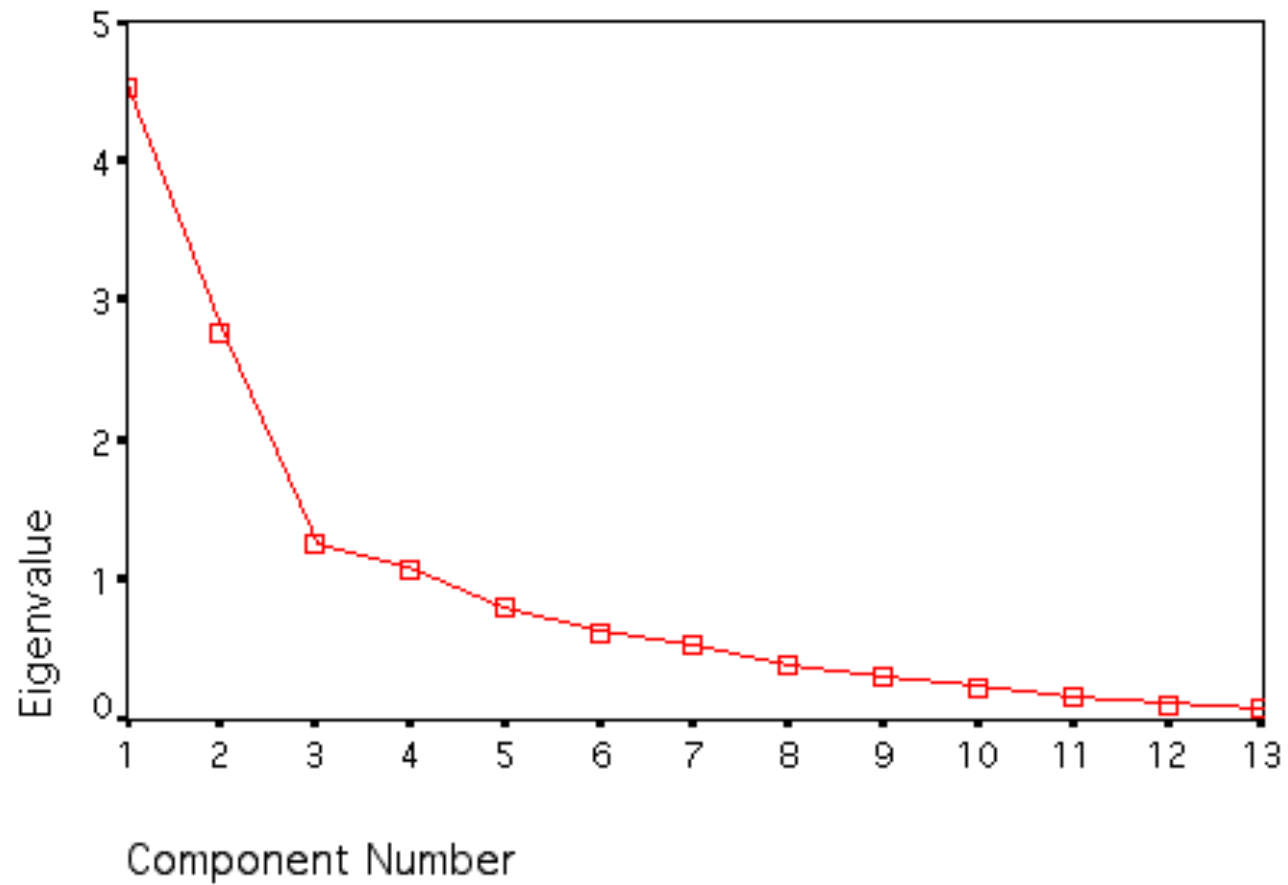
$$\text{Var}(Y_i) = \lambda_i \text{ for all } i=1, \dots, p$$

- Small $\lambda_i \Leftrightarrow$ small variance $\Leftrightarrow$ data change little in the direction of component Y_i
- The relative variance explained by each PC is given by $\lambda_i / \sum \lambda_i$

How many components to keep?

- Enough PCs to have a cumulative variance explained by the PCs that is $>50-70\%$
- **Kaiser criterion**: keep PCs with eigenvalues >1
- **Scree plot**: represents the ability of PCs to explain the variation in data

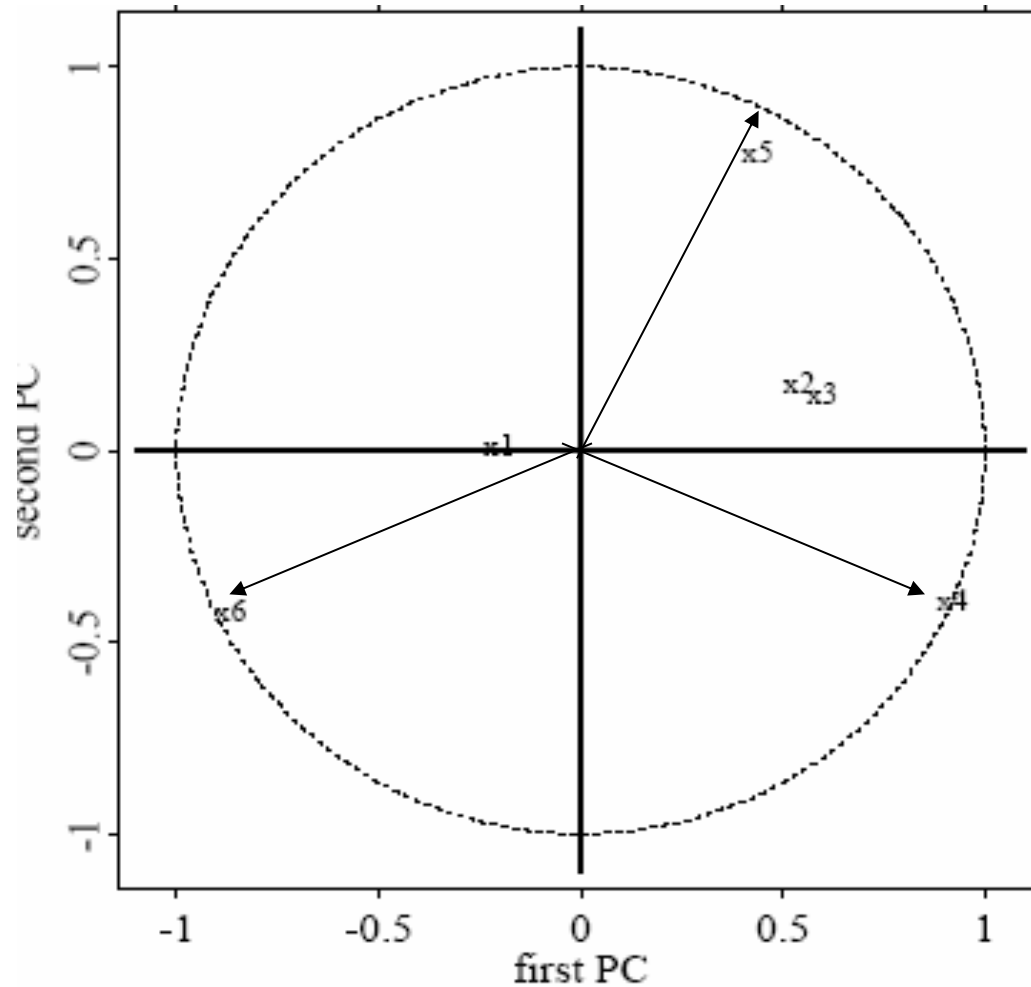
Scree Plot



Interpretation of components

- See the weights of variables in each component
- If $Y_2 = 0.89X_1 + 0.15X_2 - 0.77X_3 + 0.51X_4$
- Then X_1 and X_3 have the highest weights and so are the most important variables in the first PC
- See the correlation between variables X_i and PCs: circle of correlation

Circle of correlation



Interpreting Eigenvectors

LOADINGS

- each element of the eigenvectors represents the contribution of a given variable to a principal component
- PC_i is the dot product of $\mathbf{x}$ and $\mathbf{a}_i$

X	a1	a2	a3
Altitude	0.3842	0.0659	-0.1177
pH	-0.1159	0.1696	-0.5578
Cond	-0.2729	-0.1200	0.3636
TempSurf	0.0538	-0.2800	0.2621
Relief	-0.0765	0.3855	-0.1462
maxERht	0.0248	0.4879	0.2426
avERht	0.0599	0.4568	0.2497
%ER	0.0789	0.4223	0.2278
%VEG	0.3305	-0.2087	-0.0276
%LIT	-0.3053	0.1226	0.1145
%LOG	-0.3144	0.0402	-0.1067
%W	-0.0886	-0.0654	-0.1171
H1Moss	0.1364	-0.1262	0.4761
DistSWH	-0.3787	0.0101	0.0042
DistSW	-0.3494	-0.1283	0.1166
DistMF	0.3899	0.0586	-0.0175

PCA: *General*

From k original variables: $x_1, x_2, \dots, x_k$:

Produce k new variables: $y_1, y_2, \dots, y_k$:

$$y_1 = a_{11}x_1 + a_{12}x_2 + \dots + a_{1k}x_k$$

$$y_2 = a_{21}x_1 + a_{22}x_2 + \dots + a_{2k}x_k$$

...

$$y_k = a_{k1}x_1 + a_{k2}x_2 + \dots + a_{kk}x_k$$

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...

$$y_k = a_{k1}x_1 + a_{k2}x_2 + \dots + a_{kk}x_k$$

such that:

y_k 's are uncorrelated (orthogonal)

y_1 explains as much as possible of original variance in data set

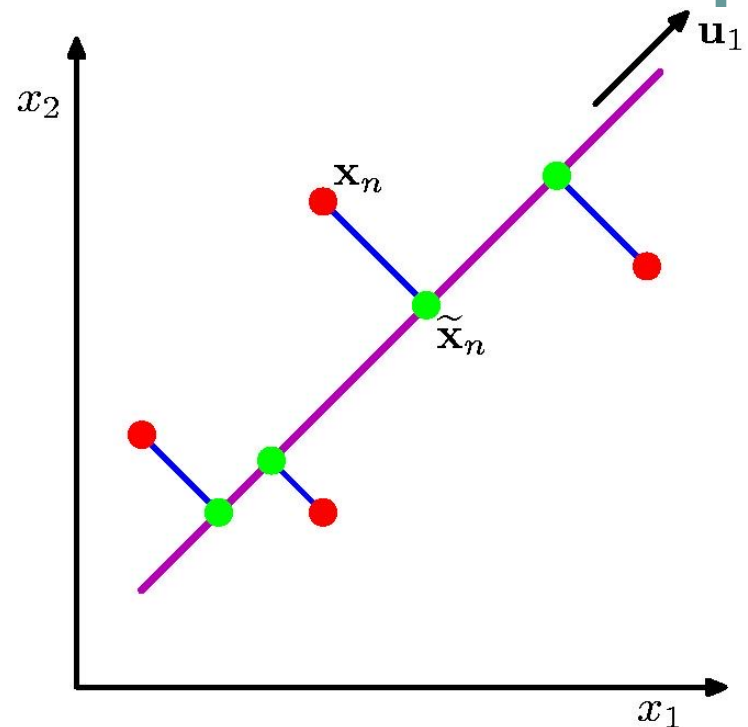
y_2 explains as much as possible of remaining variance

etc.

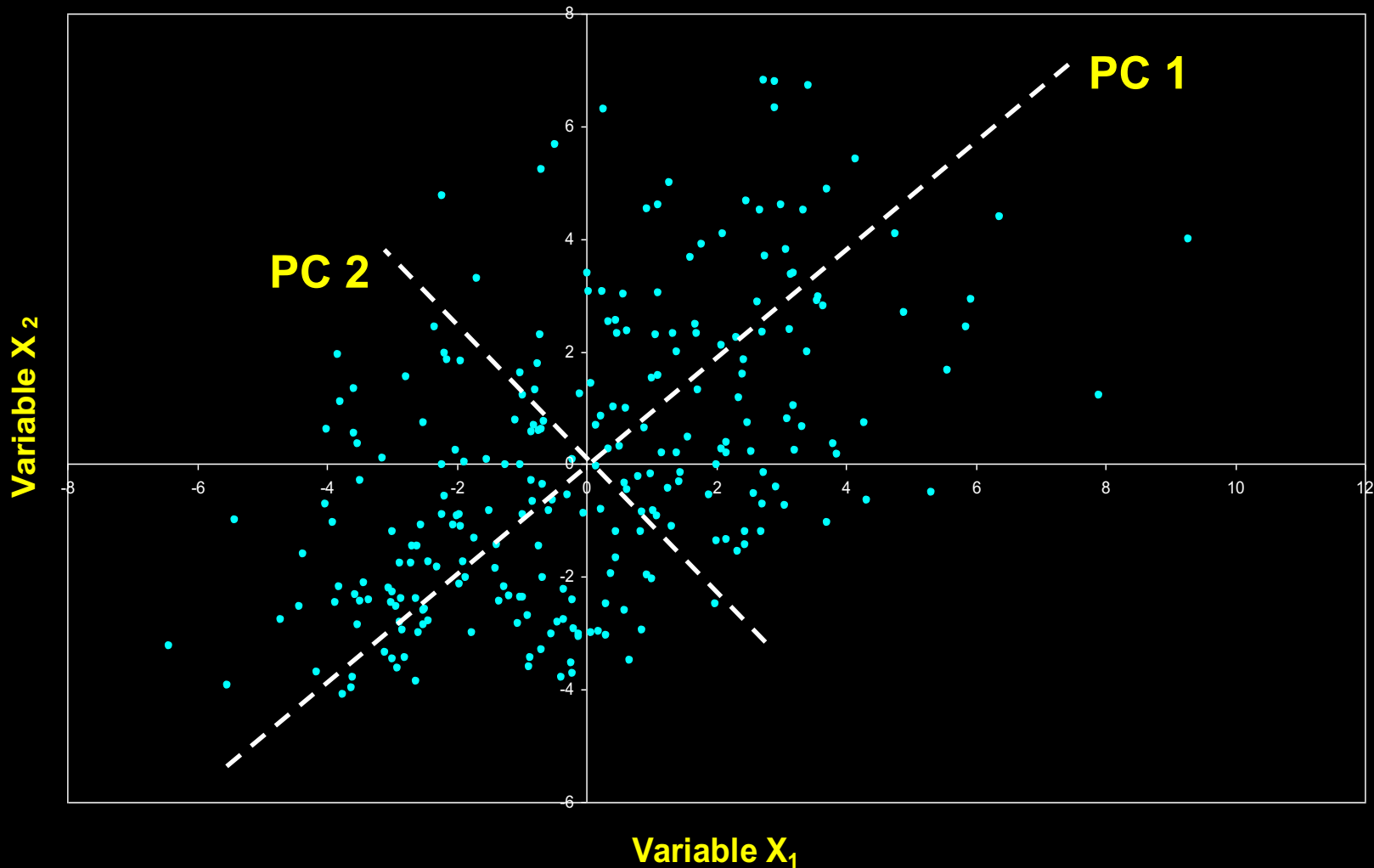
Principle Component Analysis

Orthogonal projection of data onto lower-dimension linear space that...

- maximizes variance of projected data (purple line)
- minimizes mean squared orthogonal distance between data point and projections
 - Relation with Orthogonal Regression: $PC1 = OR$



- 2D example: each principal axis is a linear combination of the original two variables
- $PC_j = a_{j1}Y_1 + a_{j2}Y_2$



Normalized (standardized) PCA

- If variables have very heterogeneous variances we standardize them
- The standardized variables X_i^* are:

$$X_i^* = (X_i - \text{mean}) / \sqrt{\text{variance}}$$

- The new variables all have the same variance (1), so each variable has the same weight.

Application of PCA in Genomics

- **PCA is useful** for finding new, more informative, uncorrelated features; it reduces dimensionality by rejecting low variance features
- **Analysis of expression data**
- Analysis of metabolomics data (Ward et al., 2003)

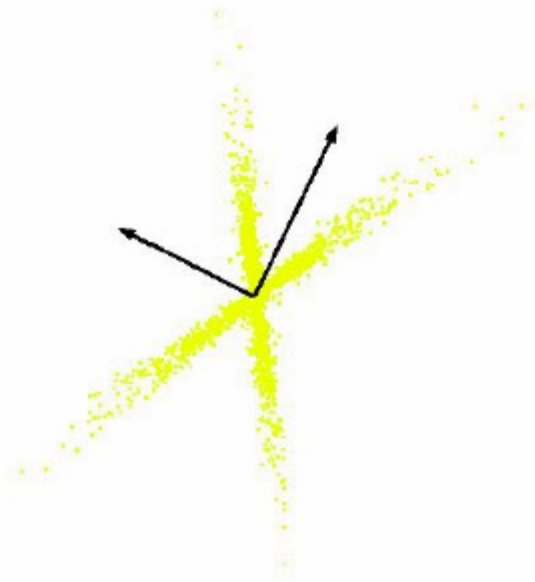
However

- PCA is only powerful if the biological question is related to the highest variance in the dataset
- If not, other techniques are more useful : Independent Component Analysis
- Introduced by Jutten in 1987

What is ICA?

- In ICA, an independence condition is optimized, which often gives more meaningful components than optimization of only the variance, as is done by PCA.
- Because of this the components of ICA are termed independent components (ICs), meaning that different ICs represent different non-overlapping information.
- For applying ICA we assume that the observed data have been determined by some unknown fundamental factors, which are independent of each other.
- By searching for components as statistically independent as possible these required factors can be detected. These fundamental factors are often termed sources and the application field is called blind source separation, BSS.

That looks like that



PCA
(orthogonal coordinate)



ICA
(non-orthogonal coordinate)

The idea behind ICA

Model:

$$x = As$$



mixing matrix

$$s = Wx$$



separating matrix

How it works?

If **independent** as well, the pdf is separable:

$$p(s_1, s_2) = p_1(s_1) p_2(s_2)$$


joint pdf marginal pdf's

which implies

$$E\{f_1(s_1) f_2(s_2)\} - E\{f_1(s_1)\} E\{f_2(s_2)\} = 0$$

for **any** functions f_1 $f_2 \Rightarrow$ useful for solving.

Rationale of ICA

- Find the components S_i that are as independent as possible in the sense of maximizing some function $F(s_1, s_2, \dots, s_k)$ that measures independence
- All ICs (except 1) should be non-Normal
- The variance of all ICs is 1
- There is no hierarchy between ICs

How to find ICs ?

- Many choices of objective function F
- Mutual information

$$MI = \int f(s_1, s_2, \dots, s_k) \text{Log} \frac{f(s_1, s_2, \dots, s_k)}{f_1(s_1) f_2(s_2) \dots f_k(s_k)}$$

- We use the kurtosis of the variables to approximate the distribution function
- The number of ICs is chosen by the user

Difference with PCA

- It is not a dimensionality reduction technique
- There is no single (exact) solution for components; uses different algorithms (in R: FastICA, PearsonICA, MLICA)
- ICs are of course uncorrelated but also as independent as possible
- Uninteresting for Normally distributed variables

Example: Lee and Batzoglou (2003)

- Microarray expression data on 7070 genes in 59 Normal human tissue samples (19 types)
- We are not interested in reducing dimension but rather in looking for genes that show tissue specific expression profile (what make tissue types different)

PCA vs ICA

- Hsiao et al (2002) applied PCA and by visual inspection observed three gene cluster of 425 genes: liver-specific, brain-specific and muscle-specific
- ICA identified more tissue-specific genes than PCA

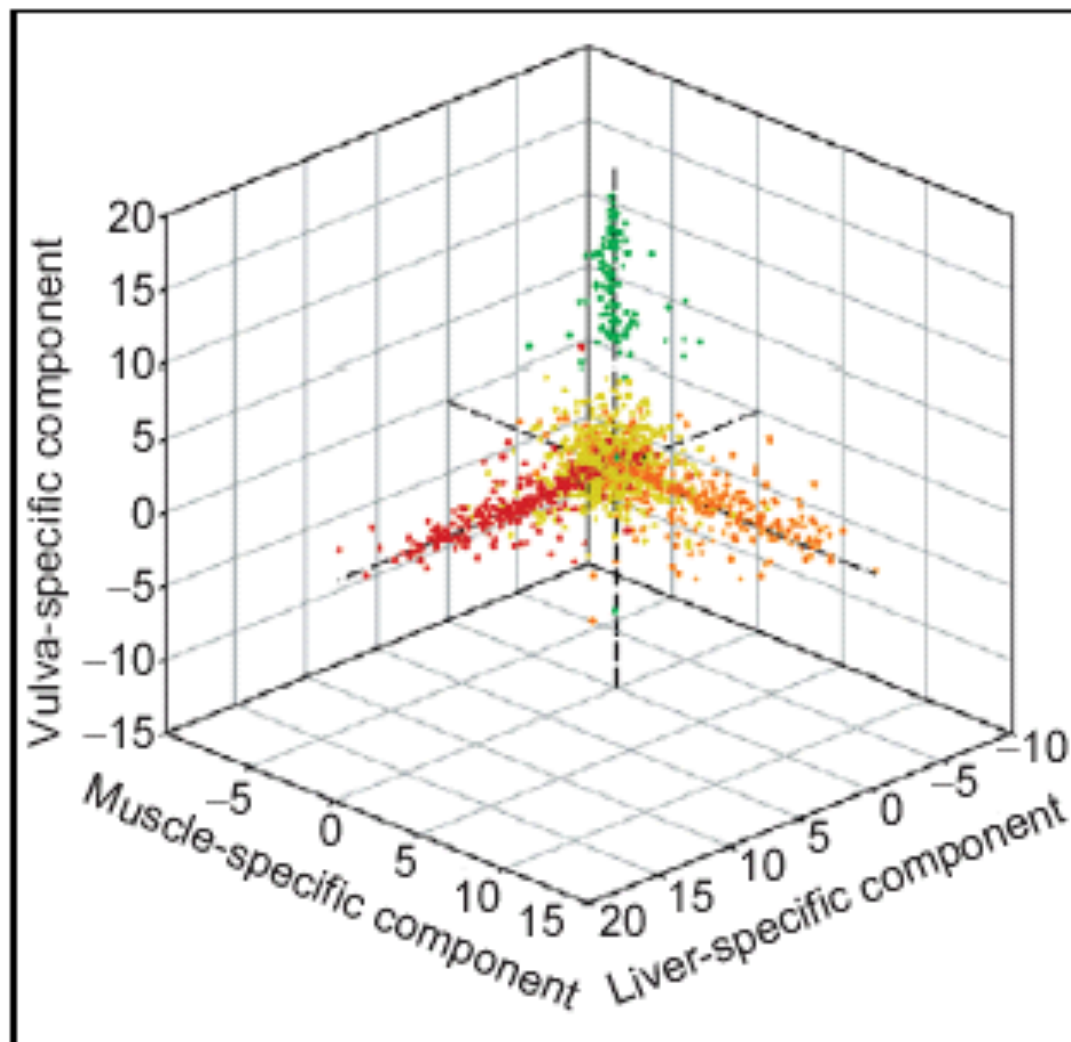
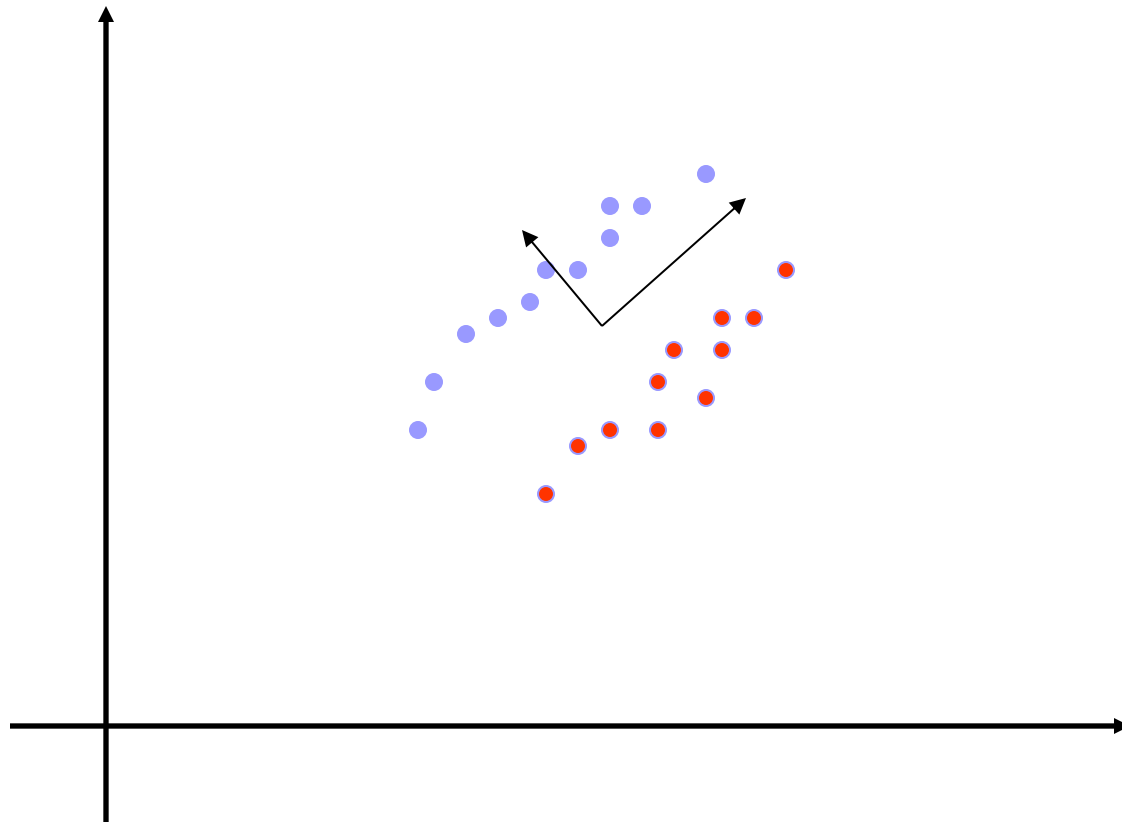


Figure 3

Three independent components of the human normal tissue data (dataset 5). Each gene is mapped to a point based on the value assigned to the gene in the 14th (x-axis), 15th (y-axis) and 55th (z-axis) independent components, which are enriched with liver-specific (red), muscle-specific (orange), and vulva-specific (green) genes, respectively. Genes not annotated as liver-, muscle- or vulva-specific are colored yellow.

More PCA Shortcomings

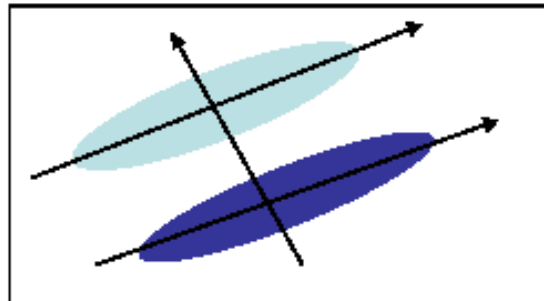
PCA, a Problematic Data Set



PCA doesn't know labels!

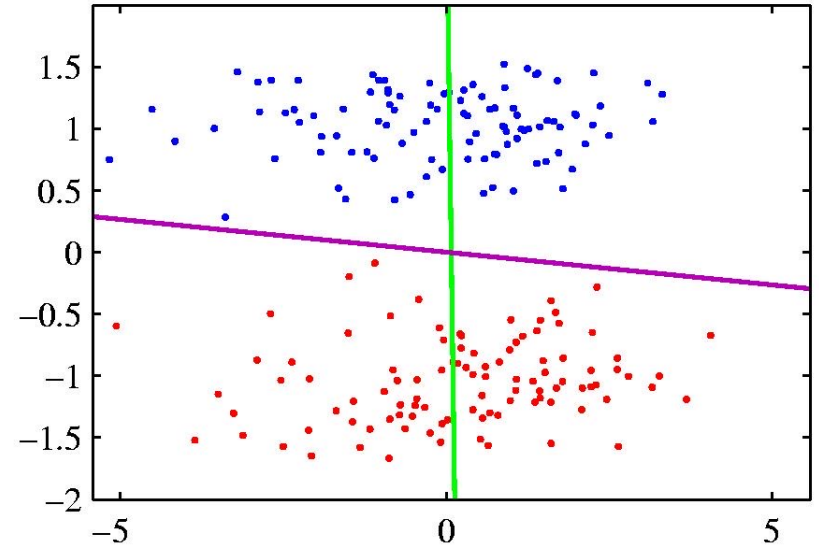
PCA and Discrimination

- PCA may not find the best directions for discriminating between two classes.
- Example: suppose the two classes have 2D Gaussian densities as ellipsoids.
- 1st eigenvector is best for representing the probabilities.
- 2nd eigenvector is best for discrimination.

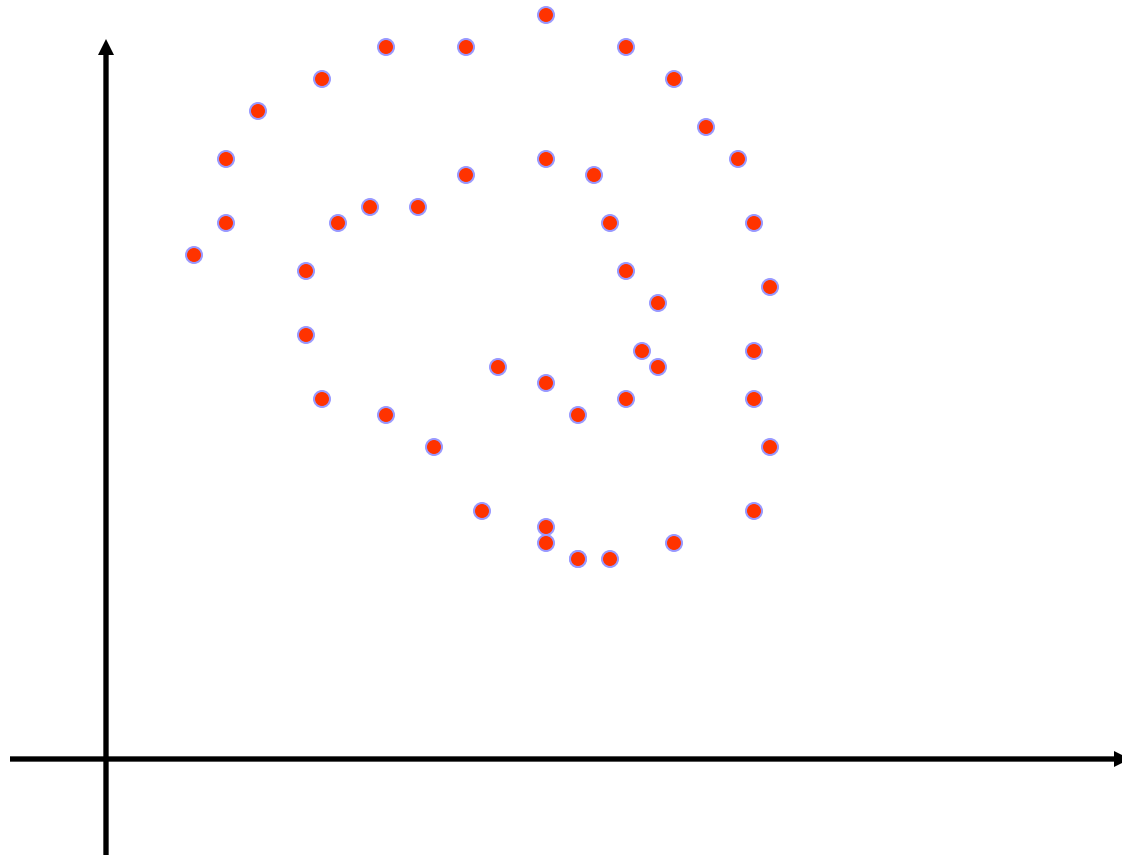


PCA vs Fisher Linear Discriminant

- PCA maximizes variance, *independent of class*
⇒ magenta
- FLD attempts to separate classes
⇒ green line



PCA, a Problematic Data Set



PCA cannot capture NON-LINEAR structure!

PCA in R

- Iris data:

<https://www.r-bloggers.com/2021/05/principal-component-analysis-pca-in-r/>

- mtcars data:

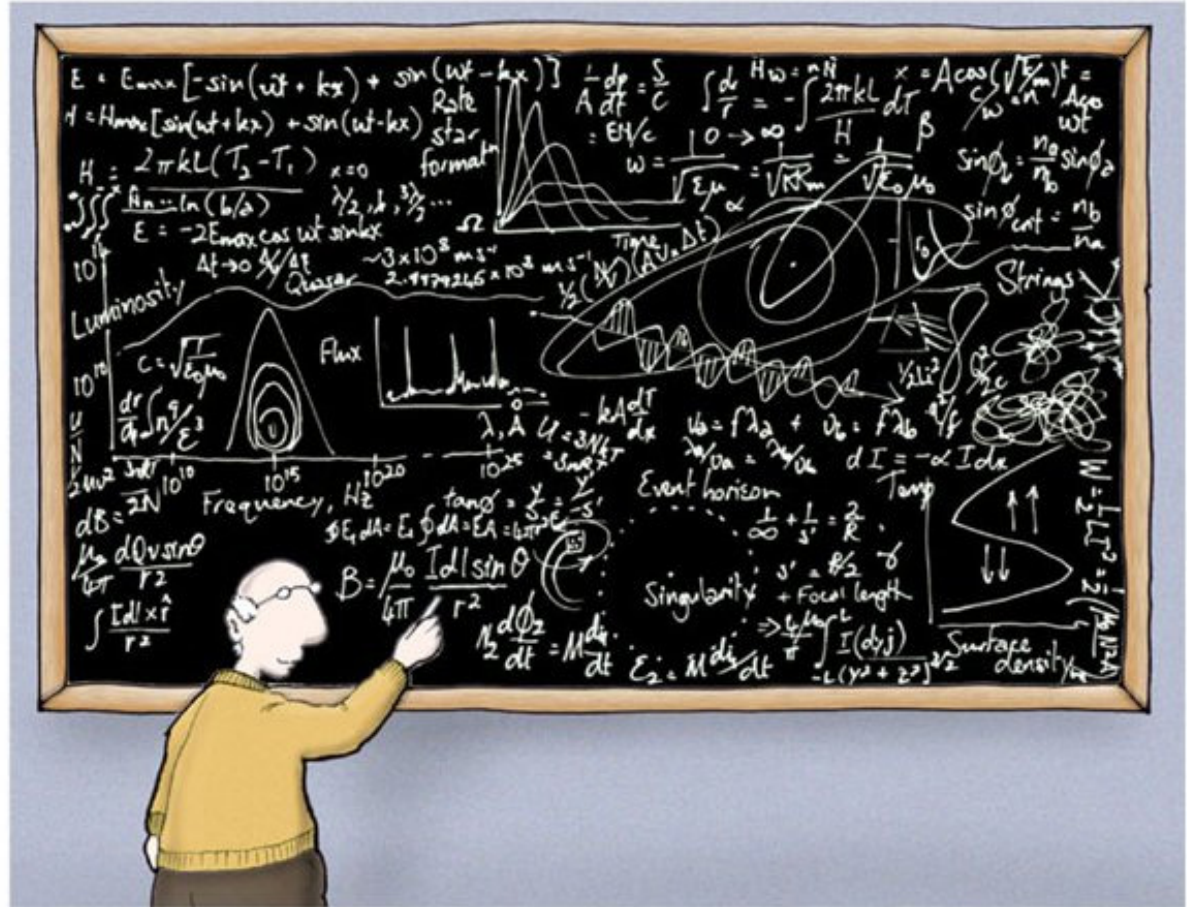
<https://www.datacamp.com/community/tutorials/pca-analysis-r>

- Breast cancer data:

<https://towardsdatascience.com/principal-component-analysis-pca-101-using-r-361f4c53a9ff>

Acknowledgement

- We thank colleagues who have kindly shared their lecture notes on the internet.



Statistics made simple